

Data Beyond Google Trends

Wikipedia pageviews and wastewater, scraped with an agent

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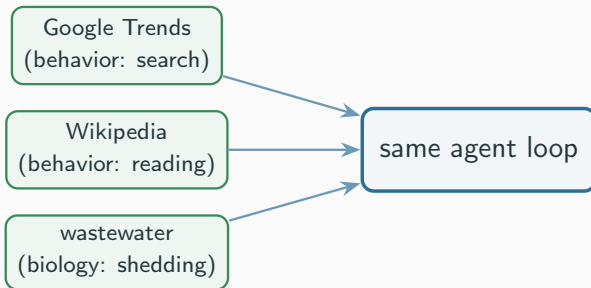
SISMID 2026 • Day 2, 9:00

Why more than search

Yesterday you got fluent with **one** stream. Google Trends is powerful but has real blind spots: attention bias, English skew, sampling instability.

Today

Add two streams that fail *differently*, so their errors do not line up: **Wikipedia pageviews** and **wastewater**. Same agent workflow, two new sources.



Wikipedia pageviews

Wastewater surveillance (CDC)

Continuing the Day 1 workflow

Wikipedia pageviews

What Wikipedia pageviews are

A free, hourly log of how many times each article was viewed, exposed through one official **Wikimedia REST API** (no key).

- Views of “Dengue fever” or “Influenza” track public attention, much like search, but from a **different behavior**: reading, not querying.
- **Stable and reproducible**: pull the same series twice and the numbers match. A welcome contrast to Google Trends.
- Deep history (2015 onward), consistent methodology, well documented.

A dependable, well-behaved stream. Good to learn the multi-source workflow on.

Language is a feature

Wikipedia is per-language, so one concept gives you **several** signals at once.

- The **Spanish** article speaks for much of Latin America, **Portuguese** for Brazil, **English** for global attention.
- This is where Wikipedia beats Google Trends: strong **country-level** and **non-English** signal, exactly where search coverage is thin.
- A natural extra time series for the Dengue exercise: overlay the Spanish and Portuguese dengue articles on the case curve.

Pick the language that matches the population you are trying to see.

Scrape it with the agent

Same five-step loop as yesterday. Only the fetch changes.

Prompt

```
Using the Wikimedia REST pageviews API (no key, set a
descriptive User-Agent), pull daily views for the
"Dengue fever" article in English, Spanish, and
Portuguese since 2016. Use all-access and all-agents,
resolve redirects to the canonical title, align to
weekly, merge on date, plot the three, and report the
date range and any gaps.
```

Then the habit: one plot, one sanity question, before it enters a model.

- **Article titles differ by language** and can be renamed; a wrong title silently returns nothing. Resolve to the canonical page and check the count is non-zero.
- **Access and agent filters matter:** all-access vs mobile/desktop, and all-agents vs user (exclude bots) change the series.
- **Attention spikes:** a news event or a linked front-page item can spike views with no change in disease, just like search.
- **Disambiguation:** “Cold” the illness vs the temperature. Prefer specific articles or topics.

Wastewater surveillance (CDC)

A fundamentally different kind of signal

Wikipedia and Google Trends measure **behavior**. Wastewater measures **biology**.

- Treatment plants are sampled for pathogen genetic material (viral RNA). It counts **shedding** by infected people, whether or not they search, test, or seek care.
- Because people shed early, it can **lead** clinical reporting, and it captures mild and asymptomatic cases that never reach a clinic.
- Publicly available from the CDC **National Wastewater Surveillance System (NWSS)**, by site and by jurisdiction.

The one stream here that does not depend on human attention at all.

Scrape it with the agent

Prompt

```
From CDC's National Wastewater Surveillance System (NWSS)
open data on data.cdc.gov, pull the weekly wastewater
viral activity level for Georgia (state aggregate, and
list the reporting sites). Tidy to date + value, plot it,
and overlay a reference such as ILI or ED visits so I can
see whether it leads. Report site coverage and any gaps.
```

The agent finds the dataset, handles the query, and returns a tidy series; you judge whether it actually leads.

Wastewater gotchas

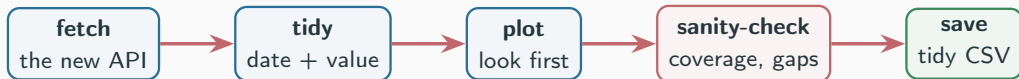
- **Coverage varies:** which plants report, and what fraction of the population they cover, changes over time and by state. Gaps are common.
- **Normalization:** raw concentration is adjusted for flow and for a fecal marker (PMMoV). Compare like with like; do not mix raw and normalized.
- **Single site is noisy:** a regional or state aggregate is far more stable than one plant.
- **Reporting lag and cadence:** it is fast, but not instant, and not every site reports every week.

Assess it honestly

The claim is early warning. Check the **lead time against ILI** yourself, as in Santillana's wastewater preprint, rather than assuming it.

Continuing the Day 1 workflow

Same loop, same habits, now three streams



- Nothing new to learn in the *workflow*: describe the outcome, let the agent fetch, then verify. Only the source changed.
- Wikipedia and CDC are well-behaved (no datacenter block), so retries and the proxy rarely come up here. Google Trends was the hard one.

You now hold a small multi-stream bundle

For your disease and place, you can now assemble, side by side:

- Google Trends (search behavior), preprocessed,
- Wikipedia pageviews (reading behavior, multi-language),
- wastewater (biological shedding),
- all against a reference such as ILI or ED visits.

This bundle is the raw material for the capstone: independent signals whose errors do not line up.

Looking ahead

- **Next (11:00):** more novel streams, human mobility (ground and air) and news alerts for emerging outbreaks.
- **This afternoon:** fold these signals into **ARGO**, alongside the disease's own history.
- **Day 3:** bring your own problem, and combine whichever streams fit it.

Two more streams down, the same loop each time. The library of signals is growing.

Questions?

Repo: `github.com/Shihao-Yang/sismid2026`